ME2 VCM — Voice Controlled Smart Device

Always-on keyword + intent recognition at the edge · no cloud round-trip

15 October 2026 · El Veena Grace Rosero · Model · Dataset · Training on Apple MPS · Validation on RPi 4

Model

Mic · 16kHz → wake CNN → log-mel 40×96

DS-CNN trunk → BiGRU(128) · 0.367 M

Intent head · 20

Slot (fine) head · 37

Dispatcher → Actuator

Raspberry Pi 4 (4gb)

Item	Value
Parameters / weights	0.367 M · ~1.6 MB ONNX (CRNN)
Compute	62.78 M FLOPs / inference
+ wake gate	0.249 M-param CNN · 1.0 MB (“Hello Jarvis”)

Validation on the Raspberry Pi 4

Item	Value
Wake / intent acc	100 % wake (5,000 clips) / 91.6 % intent (test) · 88.6 % holdout
False-accept rate	0.13 % wake FAR · 26.1 % FAR_oos @ min_conf 0.25
Latency p95 / RTF	9.36 ms classify · 4.85 ms wake (on-Pi) / RTF 0.006
Live sessions	247 wakes · 94 % recognized · 63 % acted ok · Oct 16 gate: 9/10 fired · 1 false fires
Runtime	onnxruntime (CPU) · 2 threads

Dataset

Item	Value
Source	HF airimonda/ai231-me2-voice-commands + supplementals (6 sources merged)
Hours / utts	21.0 h / 36,604 clips: 29,836 train (17.3 h) · 6,305 test · 463 holdout
Speakers	664 (speaker-disjoint splits)
Labels	20 intents · 37 fine (slot) labels · OOS class

Training on Apple MPS (local Mac)

Item	Value
Hardware	Apple M-series · torch MPS backend
Objective	CE(fine) + 0.5·CE(intent) · label smoothing 0.05
Optimiser	AdamW wd 1e-4 · lr 1e-3 cosine · batch 128
Steps / loss	60 epochs · seed 42 / 0.65 ~2.5 h wall-clock
Input	16 kHz → log-mel 40×96 · z-scored · energy-norm

To Be Submitted

Links & artefacts

Artefact	Where
GitHub repository	https://github.com/vrosero/AI231_MEx — public, MIT
Dataset location	huggingface.co/datasets/airimonda/ai231-me2-voice-commands
Training hardware	Apple M-series MPS (local Mac) · 60 epochs · seed 42
Model weights	crnn.pt + crnn.onnx (~1.6 MB) · jarvis_cnn.onnx (1.0 MB)

What the reviewer will look for

#	Item	Status
1	Repo public, one-command reproduction	✓ setup.sh + run.sh — verified on the Pi
2	Dataset licensed and citable (DOI)	✓ HF dataset → citable DOI: 10.57967/hf/10723
3	Training logs + final checkpoint committed	✓ train_crnn.jsonl · crnn.pt · onnx/
4	Pi 4 latency reproduced by the posted script	✓ perf.py bench/wake · p95 9.36 ms VCM · 4.85 ms wake (100+100)
5	Held-out test set, unseen speakers	✓ test 6,305 + holdout 463, speaker-disjoint
6	Baseline of comparable size compared	single deployed model (CRNN 0.367 M); no baseline trained yet

Demo: say “Hello Jarvis” → Jarvis responds “Yes” → say command → action